



Perception and attitudes of Brazilian livestock advisers toward cow-calf contact in dairy farms

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ABSTRACT

The prolonged cow-calf contact represents a current dilemma faced by the dairy industry, and livestock advisers play an important role in the communication between society and farmers to address the challenges of modern livestock farming. This study aimed to explore Brazilian dairy advisers' perception and attitudes regarding different prolonged cow-calf contact systems. Dairy advisers ($n = 236$) participated in an online questionnaire containing closed (multiple-choice and 5-point Likert scale) questions regarding attitudes toward prolonged cow-calf contact systems, dairy cattle welfare, and the role of advisers in communicating information between society and the farmers. Likert scale responses were grouped into a 3-point ordinal scale representing negative (score = 1), neutral (score = 2), and positive (score = 3) attitudes. Participants were randomly assigned to 1 of 4 cow-calf contact systems, using a 2×2 experimental design with full or partial contact (FullC and PartC, respectively) and full or partial time (FullT and PartT, respectively) allowance spent together. Effects of cow-calf contact systems, gender, and age on participant attitudes were analyzed using ordinal logistic regression models in RStudio. Participant gender was balanced (51.3% female and 48.7% male); 48.3% of participants were between 26 to 35 yr old, and 43% had a degree in veterinary medicine. Participants' attitudes were generally negative and did not differ across cow-calf contact systems. Also, participants showed negative attitudes toward ease of management and animal health in all cow-calf contact

systems. Despite that, participants rated animal quality of life higher in FullT-FullC and PartT-PartC compared with FullT-PartC and PartT-FullC. Participants showed more favorable attitudes toward cow-calf contact systems in small farms (mean score $\pm$ SE; 2.23 ± 0.9 and those with Zebu cattle (2.25 ± 0.8) rather than bigger farms with European breeds (1.72 ± 0.8). Men were 43% less likely than women to consider discussing these systems relevant, regardless of age. Younger male participants were less likely to be positive to cow-calf systems. In conclusion, Brazilian livestock advisers showed negative attitudes toward the prolonged cow-calf contact systems described in this study. Gender and age also influenced participants' attitudes, with younger male participants being less likely to support cow-calf contact. Our findings highlight that, even though new professionals are exposed to animal welfare concepts during their education, this alone is not sufficient to encourage the adoption of practices that address societal values.

Key words: animal care, livestock paradigms, professionals, sustainability

INTRODUCTION

Reforming animal husbandry systems is essential to align livestock farming with sustainability and societal expectations. Livestock extensionists and advisory services play an indispensable role at the frontline of the dairy industry's evolution (Cook et al., 2021). These professionals must reconcile practices that balance productivity, animal welfare, and societal demands. In this process, livestock advisers play a role in transferring knowledge and supporting farmers to adopt practices that enhance animal welfare. However, the recommendations provided by advisers are influenced by their own level of knowledge (Stutzman et al., 2019); they acknowledge gaps on certain subjects and assume that this could be one

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

of the barriers to adopting new techniques on farms (Cardoso et al., 2019; De-Sousa et al., 2023). To overcome this barrier, understanding the knowledge of advisers about actual practices is essential for the modernization of the dairy industry.

Among the dilemmas currently faced by the dairy sector (fate of male calves, De-Sousa et al., 2025; use of antibiotics, Fischer et al., 2019; dehorning, Saraceni et al., 2021), the prolonged cow-calf contact has increasingly drawn the attention of both society and researchers (Cook and von Keyserlingk, 2024; Roslindo et al., 2024; Whalin et al., 2025). Separating a calf from its mother shortly after birth is a well-established practice on dairy farms. Advocates for the early separation argue that its benefits include easier management of cows and calves, ensuring that the calf receives an adequate amount of high-quality colostrum, enabling close monitoring of the calf's health and reduction of weaning distress (Beaver et al., 2019). A variety of prolonged cow-calf contact can be implemented, but to maximize their benefits, it is essential to tailor these practices to each farm's specific context. For instance, physical contact between cows and calves can be either full or partial, differing in the opportunity for nursing, and both options can also vary in daily duration (Sirovnik et al., 2020). In full-contact systems with continuous access, calves are typically kept with the herd and can nurse from their mothers (Johnsen et al., 2016). In contrast, in full-contact systems during short periods of the day, cow-calf contact may be allowed between milkings (Bailly-Caumette et al., 2023) or at night (Barth, 2020). To minimize the effect of calf nursing on milk production and to control calf's milk intake, farmers can offer milk artificially to calves when adopting a partial-contact system, in which the calf has physical contact with the mother but is prevented from nursing (Sirovnik et al., 2020).

Although it is often considered incompatible with large-scale operations, family farms in Brazil, accounting for 78% of agricultural establishments (IBGE, 2019), commonly allow natural nursing during milking to stimulate milk letdown, as Zebu cows and their crosses (e.g., Gyr, Gyrolando) typically release less oxytocin without the presence of a calf (Santos and Bittar, 2015). Conversely, there is a movement in Brazil to select Zebu cows that are milked without calf stimulation, whereas in other countries (mainly in Europe), there is increasing interest in more natural calf-rearing practices, even within intensive production systems (Placzek et al., 2021; Cook and von Keyserlingk, 2024; Whalin et al., 2025). Brazil holds one of the world's largest cattle herds, characterized by highly diverse production systems, making it essential to understand how advisers perceive and evaluate practices that deviate from conventional norms. Thus, studying the perceptions and attitudes of Brazilian livestock advisers

regarding different cow-calf contact in dairy farms will provide valuable insights into the feasibility of these practices in intertropical zones, as these regions differ significantly from the predominant research focus on Europe.

This study aimed to explore the attitudes and perceptions of Brazilian dairy advisers toward prolonged cow-calf contact systems (CCC) and their effects on animal production by identifying barriers and potentialities. In our study, dairy advisers were considered as active professionals from rural extension services, which included veterinarians, animal scientists, and agricultural technicians with formal education in agriculture. We hypothesize that dairy advisers hold a negative attitude toward prolonged cow-calf contact, mainly driven by economic and health concerns, which may discourage recommendations for its adoption in intensive systems.

MATERIALS AND METHODS

This study was approved by the Research Ethics Committee with Human Subjects of São Paulo State University (2024/02841-7).

Recruitment of Participants

Advisers of the dairy industry were invited to participate in an online survey (developed in SurveyMonkey; <https://www.surveymonkey.com>) from April to October of 2024. We invited active professionals of rural extension services, with some professional education in agriculture and that provided a wide range of services, such as production advice and technology transfer. These included professionals from public (through government projects or funding institutions), and private companies (where farmers pay directly for advice). To avoid conflicts of interest and given that the objective of this study was not to evaluate institutions or the public and private sectors, we did not ask which company the participants worked for. Therefore, the information provided by our respondents does not represent the opinion of their institutions. The invitation was performed exclusively online by sharing the link of the survey on social media (e.g., Instagram, Facebook, and LinkedIn). Also, we sent invitations through an e-mail list of different organizations. Additionally, a snowball sampling method was used (Goodman, 1961), where each participant was invited to share the survey with their network. Conditions to participate in the survey included to be at least 18 yr old, to work as an adviser on dairy production in Brazil and, to be available and interested in voluntarily answering a questionnaire covering the general theme of "prolonged cow-calf contact systems." Before starting to answer the survey, each participant was required to read and indicate

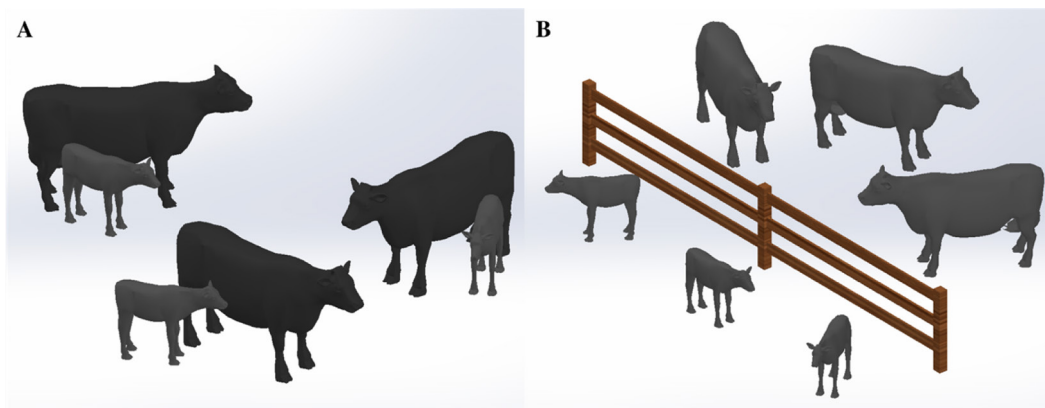


Figure 1. Schematic representation of the prolonged cow-calf contact systems. (A) Full contact (FullC): Cows could smell, lick, see, and nurse the calves, and calves could suckle. (B) Partial contact (PartC): Cows could smell, lick, and see the calves but could not nurse.

if they agreed with the terms stated in the consent form; only participants who agreed were directed to the survey.

Description of the Survey

The questionnaire (translated version to English and original version in Portuguese), available in Supplemental File S1 (see Notes), had 25 questions and was divided into 3 sections with 12 multiple-choice questions and 13 Likert scale questions of 5 points. In the first section (questions 1–6), participants were invited to answer questions regarding the welfare of dairy cattle and their role as advisers in perceiving and conveying information between society and dairy farmers.

The second section (questions 7–14) dealt with questions that covered the attitude and participants' perceptions toward prolonged CCC. For this, each participant was randomly assigned to 1 of 4 prolonged CCC using a 2×2 experimental design. The 2 factors evaluated were the type of contact (full contact [FullC] or partial contact [PartC]), and time (full time [FullT] or partial time [PartT]) allowance spent together, that also could

be full or partial. For each scenario of cow-calf contact, a short description (Table 1) and a schematic representation (Figure 1) were presented. After that, we asked the participants to answer 5-point Likert scale questions to measure their attitude toward the scenario. The third section (questions 15–25) contained sociodemographic questions.

Before initiating the data collection, we conducted a pilot validation of the questionnaire using the first 15 responses. During this phase, we carefully reviewed each participant's answers to assess the clarity, consistency, and relevance of the questions. Based on these insights, minor adjustments and refinements were made to improve the comprehensibility and appropriateness of the questionnaire items for the target audience.

Data Analysis

We received a total of 272 completed responses, 36 of which were eliminated because the respondents had no direct contact with farmers ($n = 14$), had no training in agricultural sciences ($n = 5$), or were undergraduate

Table 1. Description of the 4 hypothetical scenarios of prolonged cow-calf contact systems that differ on type of contact (FullC vs. PartC) and time allowance spent together (FullT vs. PartT)

Scenario	Abbreviation	Description
Full time and full contact	FullT-FullC	Cows and calves are raised together in the same area and are only separated during milking (morning and afternoon). In the housing area, the cows can smell, lick, see, and nurse the calf.
Full time and partial contact	FullT-PartC	Cows and calves are raised in adjacent areas separated by a fence. In the housing area, the cows can smell, lick, and see the calf, but cannot nurse.
Partial time and partial contact	PartT-PartC	Cows and calves are raised separately in different areas and only meet right after milking (morning and afternoon). In the meeting area, the cows stay for 30 min and can smell, lick, and see the calf, but cannot nurse.
Partial time and full contact	PartT-FullC	Cows and calves are raised separately in different areas and only meet during post-milking (morning and afternoon). In the meeting area, the cows stay for 30 min and can smell, lick, see, and nurse the calf.

students ($n = 17$), resulting in a final sample of 236 complete questionnaires (FullT-FullC, $n = 51$; FullT-PartC, $n = 65$; PartT-PartC, $n = 58$; and PartT-FullC, $n = 62$).

This study was characterized as exploratory; therefore, we used a convenience sample of participants (i.e., data were collected until no new information emerged; Braun et al., 2019). This ensures that the obtained data provides a rich and diverse set of responses once saturation on the proposed topic is reached (Robson; McCartan, 2016). To confirm data saturation, the survey analysis was conducted in 2 stages. First, we analyzed 100 responses across all questions. After examining the next 136 responses, no new elements were identified, indicating that data saturation had been achieved.

We used a quantitative approach to identify differences in participants' perception and attitudes toward the general theme "welfare of dairy cows and calves in prolonged cow-calf contact systems." In this context, perception refers to how individuals interpret and make sense of information based on their prior experiences, cultural background, and situational factors, whereas attitudes reflect the evaluative judgments, positive, negative, or neutral, that individuals form about a specific practice, often influencing their decisions and behaviors (Ajzen, 2005; Pickens, 2005).

The responses from Likert scale were analyzed in R language using RStudio version 2024.09.1 (R Core Team, 2024). Ordinal logistic regression models for each Likert scale question (response variable) were created to identify effects of the different prolonged CCC (FullT-FullC, FullT-PartC, PartT-PartC, and PartT-FullC), gender (female and male), and age categories (18–25, 26–35, 36–45, 46–55, and 56 or older), and interaction between age and gender on participants attitudes. Ordinal logistic regression was selected due to the ordinal structure of the dependent variable. Original responses were collected using a 5-point Likert scale and subsequently aggregated into a 3-point ordinal scale, in which lower scores (1 and 2) were grouped as negative attitudes (coded as 1), the score of 3 represented a neutral position (coded as 2), and higher scores (4 and 5) were grouped as positive attitudes (coded as 3). This recoding strategy is widely applied in survey research to enhance statistical robustness and facilitate the interpretation of ordinal data, particularly when dealing with smaller samples or uneven response distributions (Boone and Boone, 2012).

For each fixed variable and interaction, SE, and corresponding P -value were reported. Statistical significance was set at $P \leq 0.05$. Planned contrast between each of the 4 different prolonged CCC were performed to estimate the odds ratio (OR) of participants' attitudes along with 95% CI are presented in the results section. Odds ratio facilitates the interpretation of the relative effects of prolonged CCC on the likelihood of each attitude (negative,

neutral, or positive) occurring. Odds ratio values greater than 1 indicate an increase in the odds of an attitude occurring compared with the reference group (FullT-FullC), whereas OR values less than 1 indicate a decrease in the odds of that attitude occurring. For example, an OR of 1.5 indicate that the odds of the attitude occurring are 50% higher in a treatment group compared with the reference group.

Participants were asked to rank 7 practices (providing milk to the calf using an artificial teat container; providing calves with access to brushes [mechanical or static]; allowing the calf to suckle from the mother cow; raising calves in pairs or groups; raising calves in individual pens; raising calves in pasture-based systems; and separating the calf from the cow immediately after birth) from 1 (most important) to 7 (least important) in terms of their contribution to promoting welfare for dairy calves. To summarize the responses, a weighted score was calculated for each practice. First, the ranks were inverted so that higher scores indicated higher importance. Where a rank of 1 (most important) was assigned a weight of 7 and a rank of 7 (least important) was assigned a weight of 1. The weighted score for each factor was then calculated as the average of these weights across all participants.

RESULTS

The main characteristics of Brazilian livestock advisers ($n = 236$) were summarized in Table 2. Participant gender was balanced between female (51.3%) and male (48.7%), and 48.3% of participants were aged between 26 and 35 yr. Almost 43% of the participants had a degree in veterinary medicine, 52.1% of participants had up to 5 yr of professional experience assisting dairy farms, and 47.5% declared everyday frequency of interaction with farmers.

Participants attitudes did not differ ($P > 0.05$) among the cow-calf systems. Participants' attitudes were negative for all cow-calf systems, as shown by low Likert scores (means $\pm$ SE; FullT-FullC: 1.80 ± 0.12 ; FullT-PartC: 1.78 ± 0.11 ; PartT-PartC: 1.74 ± 0.12 ; PartT-FullC: 1.60 ± 0.12). There was an interaction (Figure 2) between age and gender ($P = 0.04$), indicating that younger male participants were less likely to be positive to cow-calf systems (OR: 0.57; 95% CI: 0.33–0.97). Regardless of age, male participants were 43% less likely ($P < 0.001$; 95% CI: 0.65–10.25) to consider discussing cow-calf systems as relevant in professional advisory contexts.

Overall, participants perceived that all CCC were difficult ($P > 0.05$) to manage cows and calves (means $\pm$ SE; FullT-FullC: 1.71 ± 0.08 ; FullT-PartC: 1.69 ± 0.08 ; PartT-PartC: 1.73 ± 0.08 ; PartT-FullC: 1.56 ± 0.07). Regardless of CCC and gender, age was positively associated with participants attitudes; for every category

Table 2. Profile of the Brazilian livestock advisers (n = 236) that participated in the survey of their perception and attitudes to different prolonged cow-calf contact systems

Variable	Category	n	%
Gender	Female	121	51.3
	Male	115	48.7
Age (yr)	18–25	35	14.8
	26–35	114	48.3
	36–45	49	20.8
	46–55	21	8.9
	56 or older	17	7.2
Professional education	Agronomist	32	13.5
	Veterinarian	101	42.8
	Animal scientist	86	36.4
	Other related to agricultural sciences	3	1.3
	Technical level related to agricultural sciences	11	4.7
	Other technical level	3	1.3
Postgraduate	Master's degree	57	24.2
	PhD	38	16.1
	Specialization	74	31.3
	Do not have	67	28.4
Region	North	10	4.2
	Northeast	21	8.9
	Central-West	22	9.3
	Southeast	129	54.7
	South	78	33.1
Years of experience	0–5	123	52.1
	6–10	52	22.1
	11–15	31	13.1
	16 or more	30	12.7
Kind of company	Self-employed	40	16.9
	Private company	99	41.9
	Public company	37	15.7
	Dairy industry or cooperative	31	13.2
	University	29	12.3
Frequency of interaction with farmers	Sporadically	40	16.9
	Indirectly (events or phone)	16	6.8
	Frequently	68	28.8
	Everyday	112	47.5

increase in participant age, attitude increase ($P = 0.002$; OR: 1.58; 95% CI: 1.21–2.11) on the Likert scale related to manage cows and calves.

Participants rated animals' quality of life higher ($P < 0.05$; Figure 3) in FullT-FullC (means $\pm$ SE; 2.42 ± 0.08) and PartT-PartC (2.39 ± 0.07) compared with FullT-PartC: (2.03 ± 0.07) and PartT-FullC (2.14 ± 0.07). Also, the interaction between age and gender ($P = 0.001$; OR: 0.53; 95% CI: 0.36–0.77; Figure 4) suggests that older men were 53% less likely to have a positive attitude toward animals' quality of life on CCC compared with older woman.

Animals' health was perceived ($P = 0.002$) as lower on FullT-PartC (means $\pm$ SE; 1.98 ± 0.07) compared with other CCC (FullT-FullC: 2.36 ± 0.08 ; PartT-PartC: 2.27 ± 0.08 ; PartT-FullC: 2.10 ± 0.07). Also, the interaction between age and gender ($P = 0.0003$; OR: 0.49; 95% CI: 0.34–0.72; Figure 5) suggests that older men were 49% less likely to have a positive attitude toward animals' health on CCC compared with older woman. Similar to health, animals feeling was perceived ($P = 0.001$) as

lower on FullT-PartC (2.13 ± 0.07) compared with other CCC (FullT-FullC: 2.67 ± 0.05 ; PartT-PartC: 2.48 ± 0.07 ; PartT-FullC: 2.40 ± 0.06).

General Aspects of Implementing Prolonged Cow-Calf System

Descriptively, participants' attitudes were more positive toward the cow-calf systems for small farms (means $\pm$ SE; 2.23 ± 0.9) than medium (1.62 ± 0.8) and large farms (1.34 ± 0.7). Also, participants believe that CCC are more feasible for farms with Zebu breed (2.25 ± 0.8) than farms with European breed (1.72 ± 0.8). Regarding factors that may justify the adoption of CCC (Figure 6), 64% of participants agreed with allowing natural behavior and 48% meeting consumers demands as reasons for adopting the proposed system. In contrast, only 24% of participants stated that improving cows' milk production and enhancing the health of cows' mammary glands (32%) were not seen as factors that justify adopting CCC.

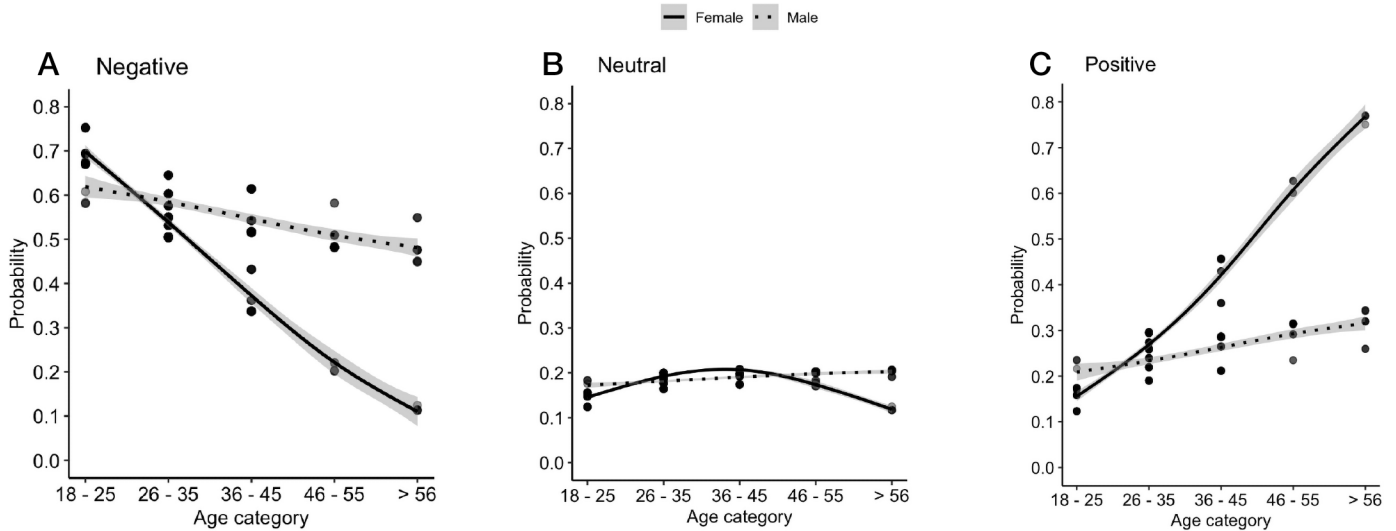


Figure 2. Probability of (A) negative, (B) neutral, and (C) positive attitudes of participants ($n = 236$) toward prolonged cow-calf contact systems. The predicted values were obtained from the ordinary regression model considering the interaction between gender (female = solid line and male = dotted line) and age category to investigate the agreement or disagreement of participants. The shaded gray areas represent the 95% CI around the predicted probabilities.

Participants' Opinions About Animal Welfare

Most participants (83%) agreed that it is important to be aware of public opinion toward the dairy industry. Furthermore, almost all participants (92%) felt responsible for helping farmers with animal welfare issues, and

71% of participants stated that they bring updated animal welfare practices to dairy farmers. Interestingly, men reported higher knowledge of animal welfare (2.76 ± 0.05) compared with women (2.65 ± 0.06). Among 7 practices that can influence calf welfare, “providing milk to the calf in an artificial teat container” received the highest

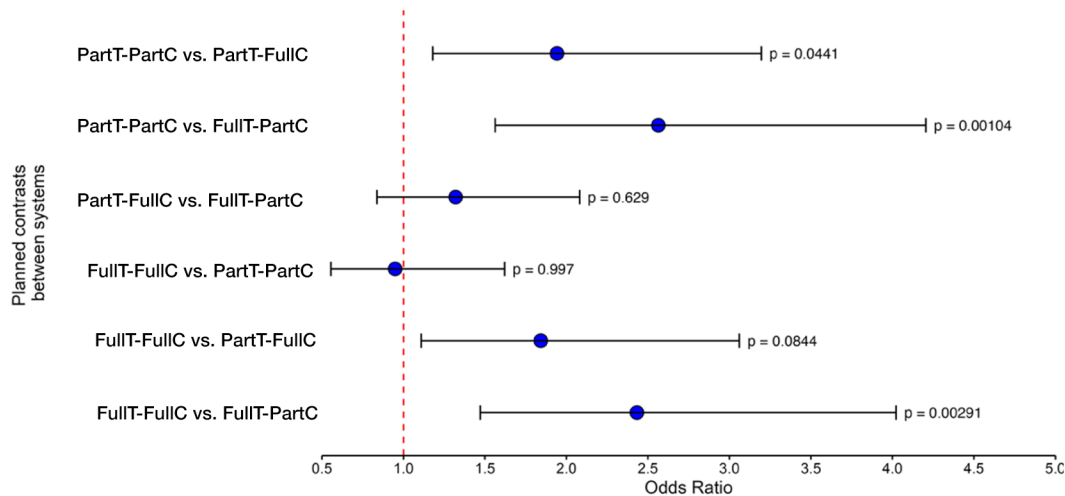


Figure 3. The estimated odds ratios and their 95% Wald confidence limits following an ordinal regression model evaluating the attitudes of participants ($n = 236$) toward animals' quality of life in planned contrasts between each of the 4 different prolonged cow-calf contact systems. The circular blue points in the graph represent the odds ratio values, and the horizontal lines extending to the left and right of each point indicate the CI for each planned contrast. The left and right endpoints of these lines represent the lower and upper bounds of the 95% Wald confidence limits, respectively. Planned contrasts with CI that do not include 1 (i.e., do not cross the vertical dashed red line) indicate rejection of the null hypothesis that the odds ratio equals 1. The shaded gray areas represent the 95% CI around the predicted probabilities.

Table 3. Percentage of Brazilian dairy advisers (n = 236) assigning ranks (1 = most important, 7 = least important) and weighted scores to practices influencing dairy calves welfare

Practice	Ranking							Weighted score
	1	2	3	4	5	6	7	
Providing milk to the calf in an artificial teat container (%)	22.6	29.3	17.6	12.9	13.3	3.9	0.4	5.22
Raising calves in pairs/groups (%)	16.0	19.9	24.6	19.9	12.9	5.5	1.2	4.85
Allowing the calf to suckle from the mother cow (%)	31.2	10.5	3.9	3.9	5.9	10.5	33.9	3.90
Providing the calf with access to brushes (mechanical or static) (%)	1.2	6.2	17.9	33.9	20.3	12.5	7.8	3.65
Raising calves in individual pens (%)	4.3	12.5	16.0	10.2	19.9	25.8	11.3	3.48
Raising calves in pasture areas (%)	2.7	11.3	16.8	13.3	20.3	27.3	8.2	3.48
Separating the calf from the cow immediately after birth (%)	21.9	10.2	3.1	5.9	7.4	14.4	37.1	3.41

weighted score, whereas “separating the calf from the cow immediately after birth” received the lowest score (Table 3).

DISCUSSION

We confirmed our hypothesis that Brazilian dairy advisers hold negative attitudes toward prolonged CCC. Concerns expressed regarding animal breed, health, management practices, and farm size may help explain their negative attitudes. Participants stated that CCC is feasible for small farms and Zebu breeds, but not for medium to large farms and European breeds. Cow-calf contact systems have been tested and implemented in European farms using European breeds, demonstrating their feasibility beyond small-scale operations (Eriksson et al., 2022). Surveys conducted with farmers from various countries (Germany: Hautzinger et al., 2025; Norway:

Johanssen et al., 2023; other European countries: Eriksson et al., 2022; New Zealand: Neave et al., 2022) indicate that CCC is implemented across a range of common husbandry systems. Indeed, some farmers from those studies perceived housing constraints as a challenge to practicing CCC; nonetheless, they expressed positive emotions toward the practice.

Gender and age shaped the attitudes of survey participants. Overall, younger participants were less likely to support CCC and exhibited more negative attitudes toward animal management in this system. We suggest that younger or less experienced advisers often have limited hands-on experience and, having recently graduated, may not have received specialized training in specific areas such as animal management and welfare. Consequently, their perspective may have been shaped by exposure to professional discourses focused primarily on farm profitability and productivity. Maybe

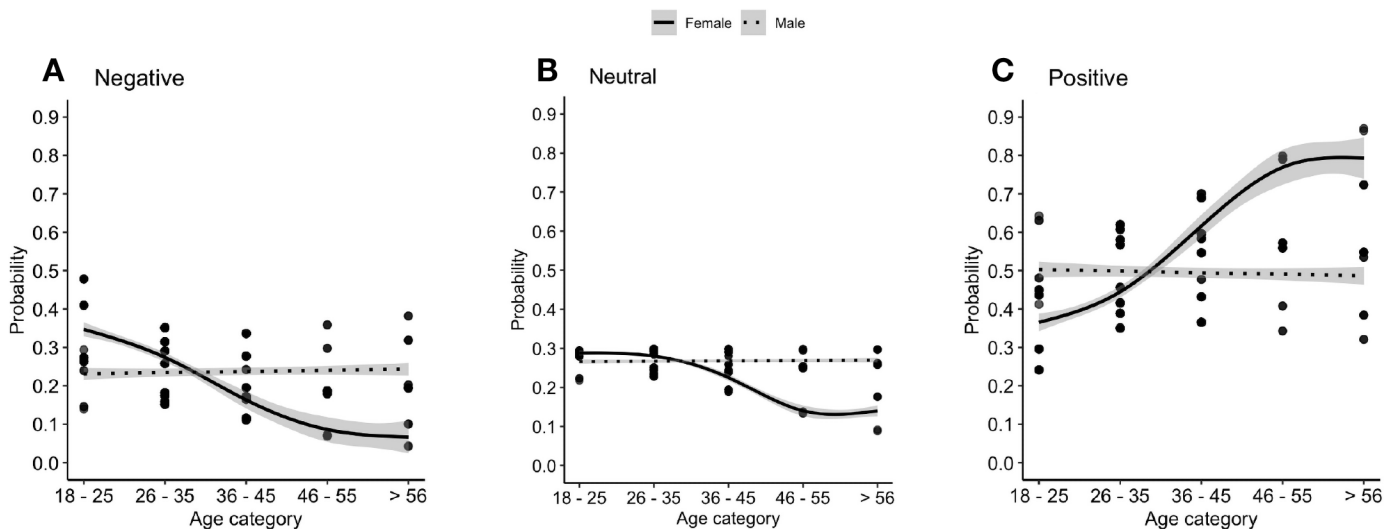


Figure 4. Probability of (A) negative, (B) neutral, and (C) positive attitudes of participants (n = 236) toward animals' quality of life on prolonged cow-calf contact systems. The predicted values were obtained from the ordinary regression model considering the interaction between gender (female = solid line and male = dotted line) and age category to investigate the agreement or disagreement of participants. The shaded gray areas represent the 95% CI around the predicted probabilities.

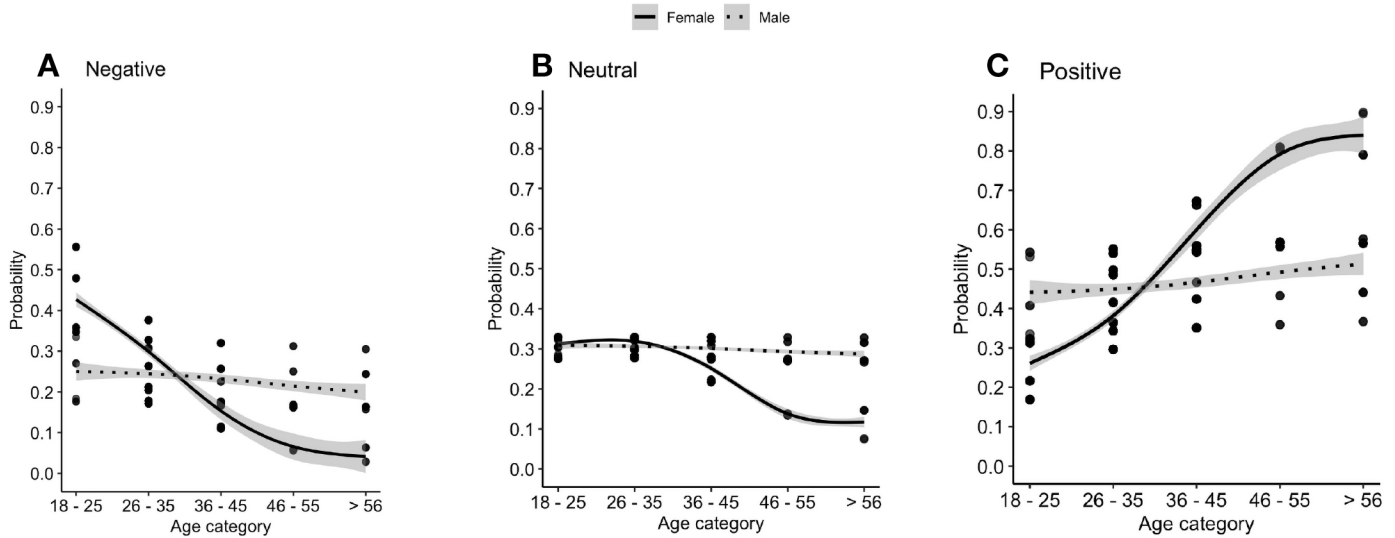


Figure 5. Probability of (A) negative, (B) neutral, and (C) positive attitudes of participants ($n = 236$) toward animals' health on prolonged cow-calf contact systems. The predicted values were obtained from the ordinary regression model considering the interaction between gender (female = solid line and male = dotted line) and age category to investigate the agreement or disagreement of participants. The shaded gray areas represent the 95% CI around the predicted probabilities.

the actual teaching systems are forming professionals who only receive conventional information (e.g., individual housing, gene editing for profit, zero-grazing systems, and so on) and are out of the advances of society's desire for modern livestock farming. The dilemma of the cow-calf contact is just one example of future transformations, other practices toward which livestock advisers have expressed negative attitudes are the fate of male calves (De-Sousa et al., 2025) and the adoption of environmentally sustainable practices (e.g., silvopastoral systems; De-Sousa et al., 2023). In the current global scenario of information overload and

rapid technological advancement, we need professionals who move forward, capable of analyzing problems, developing solutions, and applying them in real-world situations (FAO, 2019). In this process, the economic aspect of production systems cannot be ignored, but we also cannot close our eyes to the evolution of animal welfare concepts and the growing pressure from society. Among the desired skills for professionals working in extension and rural advisory services are critical thinking, problem-solving, leadership and the ability to inspire and motivate (Blum et al., 2020). Therefore, it is essential to train professionals capable of critically

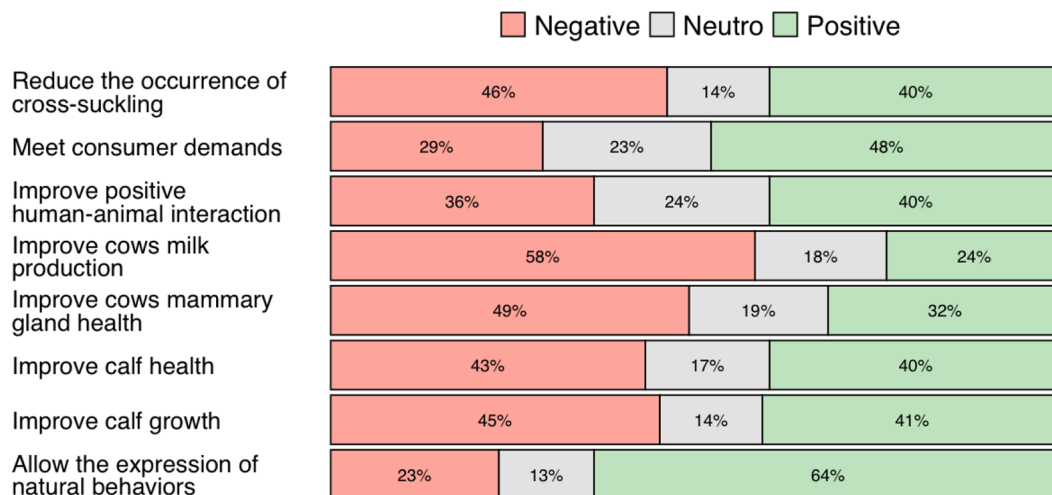


Figure 6. Agreement of participants ($n = 236$) with reasons that may justify the adoption of prolonged cow-calf contact systems.

analyzing these issues and working on the frontlines of transformations in the dairy industry.

Among older participants, men were more negative than women, particularly regarding aspects such as potential effects of CCC on animal welfare and health. Moreover, regardless of age, men were less likely to consider discussions about this practice relevant. Differences in perception between genders and ages may also be influenced by broader societal roles and exposure to animal welfare discussions, as men and women often prioritize different aspects of animal care and production ethics (Randler et al., 2021; Carnovale et al., 2022). One possible explanation is that women may draw parallels between human motherhood and the cow-calf bond, potentially attributing greater emotional significance to prolonged cow-calf contact. However, our study did not assess motherhood experiences, parental status, or empathy-related traits. Future research should investigate whether factors such as motherhood roles, empathic orientation, or anthropomorphic tendencies help explain gender-related differences in attitudes toward CCC.

The higher self-assessed knowledge on animal welfare by men surveyed was not translated into more positive attitudes, suggesting that perceptions of feasibility and management challenges may override theoretical understanding of animal welfare concepts. Additionally, the recognition that cow-calf systems promote the expression of natural behavior and are a way to meet consumer demands for animal welfare contradicted the negative attitude to all presented CCC. This seems anchored in participants' perceptions of these systems as unfeasible or inconsistent with the practical realities of farm operations. Such findings highlight a tension between theoretical animal welfare principles and real-world applications. For instance, a study with dairy farmers in New Zealand identified significant barriers to adopting prolonged CCC, including concerns about animal welfare, increased labor demands, and the compatibility of these systems with existing management practices (Neave et al., 2022). Conversely, farmers who had already implemented CCC on their own farms reported that their motivation for adopting such systems was driven by perceived benefits for animal welfare, calf health, and development (Hautzinger et al., 2025). Therefore, we suggest that future studies should assess advisers' knowledge of specific technical aspects of practices that promote prolonged cow-calf contact.

Interestingly, participants in our study rated the animal's quality of life higher in consistent systems, in which time and contact were applied at the same level (FullT-FullC or PartT-PartC), compared with mixed systems, in which time and contact were applied at different levels (PartT-FullC or FullT-PartC). This suggests that consistency in management is perceived as more beneficial for

animal welfare. In contrast, systems that combine full contact to specific times, or prevent full contact when cow and calf are kept in proximity, may not be sufficient to meet the emotional needs of the animals. Full contact combined with full time (23 h of contact/d) systems seem more likely to satisfy maternal motivation than full contact at partial time (10 h of contact/d; Jensen et al., 2024). Regarding type of contact, cows in systems that allow full contact spend more time performing affiliative behaviors toward their calf than partial-contact cows (Wenker et al., 2022). Also, the motivation to access the calf is lower in cows that are not allowed to suckle compared with those that are (Wenker et al., 2020). Further animals' emotions need; systems where time of contact is intermittent (PartT-FullC) or contact is partial (FullT-PartC) probably were related to more workload. There are conflicting views regarding workload among farmers that adopt or not adopt CCC, which is likely related to context and individual preferences (see review: Whalin et al., 2025). Although farmers practicing cow-calf contact report greater job satisfaction, partly because they spend less time feeding calves (Eriksson et al., 2022; Hansen et al., 2023; Johanssen et al., 2023), farmers using conventional cow-calf systems view the increased workload and potential health issues as major barriers to adopting this system (Neave et al., 2022). This perception, shaped by previous experiences, reinforces the idea that many advisers may be caught in the same loop: having entered the labor market with conventional systems in mind, they tend to reproduce this model and therefore perceive potential barriers without necessarily considering practical contexts or alternative management possibilities.

Participants did not perceive improved mammary gland health as a reason to adopt CCC. It is known that CCC does not compromise the mammary gland health, although cows that are allowed to nurse their calves tend to produce milk with slightly lower fat content (Nicolao et al., 2022; Wenker et al., 2022). Additionally, the cows' milk production was perceived as a negative aspect to support the adoption of CCC. Studies have shown that machine-obtained milk yield is indeed influenced by the type of contact system adopted. For example, systems that allow unrestricted calf suckling throughout the day are often associated with lower machine milk yield compared with non-CCC systems (Barth, 2020; Nicolao et al., 2022; van Zyl et al., 2025). Another aspect that may have influenced advisers' attitudes was the nutritional management of calves, which was not explicitly addressed in our survey. Studies highlight that the availability of supplemental milk sources (e.g., bottle or automated milk feeder) alongside CCC can reduce calves' dependency on maternal milk (Johnsen et al., 2015, 2018). However, the nutritional plan for calves in prolonged cow-calf contact is an unexplored topic (Whalin et al., 2025). If we

had introduced scenarios with guaranteed calf nutrition through alternative sources, the participants' concerns regarding milk yield loss and calf growth might have been mitigated.

Although our study provides valuable insights into the perception and attitudes of Brazilian livestock advisers toward cow-calf contact in dairy farms, some caution is required in extrapolating our findings. The convenience sample used in this study should not be considered fully representative of Brazilian livestock advisers due to the diversity of professionals in this field. However, our findings reflect the perspectives of a group with varied backgrounds in agricultural education, age, and years of experience in private and public sector advisory roles. The methodology assumed that participants had some level of formal education in dairy farming, which was confirmed by the participant profile data. Additionally, we considered whether advisers visited the farms they provided guidance to, supporting the assumption that they had familiarity with the dairy farm.

Our study highlights the need to bridge the gap between theoretical welfare principles and their practical application. This raises an important question about the effectiveness of the animal welfare training in changing perceptions and practices in the dairy industry. Although new professionals are exposed to animal welfare concepts during their education, we believe this is not enough to encourage the adoption of new practices that address societal values (Broom, 2005; Hazel et al., 2011). To overcome it, encouraging dialog among advisers, farmers, and society, integrating scientific evidence with on-farm experiences, developing feasible management strategies and encouraging continuous professional training are essential to ensure the effective application of new practices in modern livestock farming (Wieland et al., 2021). However, continuous training alone will not be effective unless it is accompanied by new teaching strategies (e.g., artificial intelligence use: Almasri, 2024; technology: Szymkowiak et al., 2021) and critical thinking (Uribe-Enciso et al., 2017).

CONCLUSIONS

Brazilian livestock advisers expressed negative attitudes toward cow-calf contact in dairy farms, regardless of the system design. Gender and age influenced participants' attitudes, with younger male participants less likely to support CCC. Although male participants self-reported high knowledge of animal welfare compared with women, they had negative attitude toward animals' health and quality of life on CCC, contradicting finding in the literature. Our results highlight the need for targeted educational initiatives and further research to address perceived barriers, promoting informed discussions

and evidence-based decision-making regarding CCC in Brazilian dairy farming.

NOTES

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Nonstandard abbreviations used: CCC = cow-calf contact system; FullC = full contact; FullT = full time; OR = odds ratio; PartC = partial contact; PartT = partial time.

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